Trace and Shear Modes of a Three-Orientation Phase Response: An Effective Scalar Framework with a Newtonian Limit

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Abstract

We build an effective scalar order-parameter framework in which a mass distribution induces static phase shifts among three internal orientations of a hypothetical microscopic lattice. The static response separates by linear algebra into an isotropic trace mode $\boldsymbol{v(r)}$ and traceless shear modes, the spread of which we encode in an ellipticity $\boldsymbol{\mathcal{E}(r)}$ that measures relative coherence loss. By construction, the trace mode satisfies a Poisson equation. When combined with a matter–lattice energy functional restricted by three-fold vacuum symmetry, the framework reproduces the Newtonian inverse-square law, with $\boldsymbol{G = \mu\lambda\alpha}$ in terms of three effective Planck-scale parameters. The framework is explicitly effective: the Green-function kernel and energy functional are inputs, and the Newtonian limit is recovered, not derived. The novel content lies in the shear sector. For a minimal anisotropy configuration, we predict a correlated anisotropic phase response with ratio $\boldsymbol{v/\mathcal{E} = 1/(\sqrt{6}\,\epsilon)}$. Under a phenomenological source-locked spatial projection, precision tests of frequency isotropy bound the parameter ratio $\boldsymbol{|\epsilon/\lambda| \lesssim 10^{-10}}$. The framework predicts this signal scales linearly with $\boldsymbol{|\Phi_{grav}|/c^2}$ and is sourced by local gravitating mass distributions. We identify three open problems: a microscopic Hamiltonian realizing the trace-mode coupling, the Planck-scale origin of asymmetry parameters, and a dynamical mechanism for coherence loss. The framework is offered as a kinematic building block rather than a candidate theory of gravity.

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1 Introduction

Several independent research programs share the working hypothesis that gravitational phenomena may be the effective long-wavelength expression of more fundamental degrees of freedom. Sakharov's induced-gravity proposal [1] suggested that the Einstein–Hilbert action could arise as a one-loop effect from matter fluctuations. Jacobson [2] derived the Einstein field equations from local thermodynamic relations across causal horizons. Verlinde formulated gravity as an entropic force [3] and later proposed an emergent-gravity scenario in de Sitter space [4]. Padmanabhan [5] developed a thermodynamic framework in which the gravitational equations of motion reflect the equipartition of degrees of freedom associated with spacetime horizons. Hu [6] pursued a condensate-of-spacetime picture motivated by analogies with critical phenomena.

A complementary set of programs constructs gravity as the continuum limit of discrete or algebraic substrates. Loop quantum gravity [7, 8] provides a kinematic Hilbert space of quantum geometry from spin networks. Causal dynamical triangulations [9] produce four-dimensional geometries from sums over piecewise-flat manifolds. Causal set theory [10] encodes spacetime as a partial order of discrete events. Group field theory [11] unifies several of these approaches in an algebraic framework. Analogue-gravity proposals [12, 13] reproduce gravitational kinematics in condensed-matter systems through effective metric structures.

A complementary line of investigation, situated at the intersection of gravitation and quantum theory, suggests that gravity may play a fundamental role in the breakdown of quantum coherence. Within the Diósi–Penrose framework, gravitational self-energy is argued to induce an intrinsic instability of macroscopic quantum superpositions, leading to an effective state-reduction mechanism [14, 15]. The phenomenological and experimental implications of such gravity-induced decoherence models have been extensively reviewed in [16].

The present paper does not propose a microscopic theory but we work in the spirit of an effective framework: a static phase-response ansatz for a three-orientation lattice is the input, and the structural consequences are derived. The Newtonian inverse-square law is recovered in the static limit, but it is recovered *by construction*, since the assumed Green-function response kernel already encodes the Laplacian structure. The genuine content of the framework is therefore not the recovery of Newtonian gravity, which serves only as a consistency check, but the decomposition of the phase response into an isotropic trace mode and traceless shear modes. The trace mode is the field $v(\mathbf{r})$, to which the test-mass energy couples at leading order; the shear modes are encoded in an ellipticity $\mathcal{E}(\mathbf{r})$ that measures relative coherence loss across the three orientations. The framework predicts a fixed relation between the two sectors: The trace mode $v(\mathbf{r})$ and the ellipticity $\mathcal{E}(\mathbf{r})$ are simultaneous outputs of the same phase-response ansatz, so any local gravitational potential coexists with a specific anisotropic shear signal. For a minimal anisotropy configuration the ratio $v/\mathcal{E}$ is model-independent. This coexistence is the testable structural content of the framework.

We present the framework as a kinematic building block. The microscopic completion (an explicit Hamiltonian whose effective energy functional reproduces the trace-mode coupling, a Planck-scale origin for the anisotropy parameters, and a

dynamical realization of coherence loss) is an open project. Section 2 introduces the phase-response ansatz. Section 3 performs the trace–shear decomposition. Section 4 derives the Newtonian limit from a symmetry-constrained energy functional. Section 5 maps the anisotropic phase response onto observable Lorentz-violating signatures, derives bounds from existing experimental data, and proposes a direct scaling test. Section 6 states the scope of the framework and identifies open problems.

2 The Phase-Response Ansatz

Let us consider a hypothetical microscopic lattice whose elementary cells possess three internal orientations indexed as $i \in \{1, 2, 3\}$. In the vacuum configuration these orientations are phase-locked with relative offsets of $2\pi(i-1)/3$. At the present stage, the construction is intended only as an effective parametrization, and no specific microscopic realization is assumed. The choice of three orientations therefore should be viewed simply as the minimal symmetric structure capable of supporting the trace–shear decomposition introduced later in the paper. Whether this structure ultimately originates from tetrahedral lattice variables, internal gauge-like sectors, or some other microscopic organization is left open for future work.

Postulate 1 (Static phase response to mass) Let $\rho_m(\mathbf{r})$ be a static mass density. The phase of the i-th orientation at position $\mathbf{r}$ is

$$\theta_i(\mathbf{r}, t) \;=\; \omega(\mathbf{r})\, t \;+\; \frac{2\pi(i-1)}{3} \;+\; \delta\theta_i(\mathbf{r}), \tag{1}$$

with the orientation-dependent static perturbation

$$\delta\theta_i(\mathbf{r}) \;=\; -\alpha \int \frac{\rho_m(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\,(1+\epsilon_i)\, d^3r', \tag{2}$$

and a common-mode frequency shift

$$\omega(\mathbf{r}) \;=\; \omega_0\left(1 - \zeta_P \int \frac{\rho_m(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, d^3r'\right). \tag{3}$$

The parameters $\alpha, \zeta_P \in \mathbb{R}_{>0}$ have dimension LM^{-1}. The asymmetry parameters $\epsilon_i \in \mathbb{R}$ satisfy $|\epsilon_i| \ll 1$ and the tracelessness condition

$$\sum_{i=1}^{3} \epsilon_i \;=\; 0. \tag{4}$$

Several features of this ansatz deserve comment. First, the kernel $1/|\mathbf{r}-\mathbf{r}'|$ is simply the Green function of the three-dimensional Laplacian. Its presence in equation (2) is an *input* of the effective framework: it fixes the spatial response to be the leading-order long-wavelength behavior of a system whose fluctuations are governed by a quadratic gradient term. A microscopic derivation of this kernel from the continuum limit of a specific lattice dynamics is outside the scope of the present paper.

Second, the response is linear in ρ_m. This is the leading term in a weak-source expansion and is the only regime in which the framework is currently formulated. The

domain of validity is $|\delta\theta_i(\mathbf{r})| \ll 1$ for all i at all positions of interest, which for a point source M at distance r reads

$$\frac{\alpha M}{r} \ll 1. \tag{5}$$

Third, the tracelessness condition (4) is the leading-order expression of macroscopic rotational isotropy of the static response. Microscopically, we know that the local stress-energy tensor $T^{\mu\nu}$ contains directional information, and a coupling to three internal orientations will in general be anisotropic at fixed microscopic configuration. Equation (4) imposes that no net direction is preferred in the average over the underlying microscopic states; it does not require $\epsilon_i = 0$.

Fourth, the common-mode frequency shift $\omega(\mathbf{r})$ has the same functional form as the weak-field gravitational redshift. The parameter ζ_P is a free Planck-scale coupling at this stage of the construction; its identification with G/c^2 is recovered as a consistency condition in Section 4.

Table 1 Dimensions of the principal effective quantities. M, L, T denote mass, length, time. Phases, v, $\mathcal{E}$, ϵ_i, and λ are dimensionless.

Quantity	Dimension	Meaning
ρ_m	ML^{-3}	mass density
α	LM^{-1}	phase–mass coupling
ζ_P	LM^{-1}	frequency–mass coupling
v	1	trace phase mode
$\mathcal{E}$	1	ellipticity / shear-mode spread
μ	L^2T^{-2}	energy per unit mass per unit dimensionality
λ	1	trace-to-dimensionality conversion
β_0	L^2T^{-2}	first-order matter–lattice coupling
$G = \mu\lambda\alpha$	$L^3M^{-1}T^{-2}$	Newton constant

3 The Trace–Shear Decomposition

3.1 The Scalar Trace Component

The set $\{\delta\theta_i\}$ naturally decomposes into an isotropic component given by the arithmetic mean together with the traceless components determined by the relative differences between orientations. We define the scalar trace mode as minus the arithmetic mean:

$$v(\mathbf{r}) \;\equiv\; -\frac{1}{3}\sum_{i=1}^{3}\delta\theta_i(\mathbf{r}). \tag{6}$$

The sign is fixed by the convention that $v > 0$ wherever mass is present.

Substituting (2) and (4),

$$v(\mathbf{r}) \;=\; \alpha\int \frac{\rho_m(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\,d^3r'. \tag{7}$$

The asymmetry parameters ϵ_i drop out by the tracelessness condition. The scalar trace mode is therefore an isotropic, model-independent function of the source.

Proposition 1 (Poisson equation for the trace mode) *For any mass distribution ρ_m for which (7) exists, the trace mode satisfies*

$$\nabla^2 v(\mathbf{r}) \;=\; -4\pi\alpha\,\rho_m(\mathbf{r}). \tag{8}$$

Proof Apply $\nabla^2_{\mathbf{r}}$ to (7) under the integral, and use the distributional identity $\nabla^2|\mathbf{r}-\mathbf{r}'|^{-1} = -4\pi\delta^3(\mathbf{r}-\mathbf{r}')$, which holds in the sense of distributions on $\mathbb{R}^3$ [23, 24]. □

Remark 1 Proposition 1 follows directly as a consequence of the Green-function structure of the response kernel (2). It confirms that the integral and differential representations of the framework agree with each other rather than providing a microscopic derivation of Poisson's equation itself. A microscopic derivation would require showing that the long-wavelength continuum limit of a specific lattice Hamiltonian produces precisely this response.

Remark 2 (A note on terminology) The trace mode $v(\mathbf{r})$ admits a natural geometric interpretation, anticipating the spatial tetrad projection introduced in Section 5.2. If the three orientations are viewed as the directional links of a microscopic lattice cell, the traceless shear modes describe shape-distorting twists among those links, while the trace component represents a uniform volumetric dilation of the cell — a shift in the interstitial bulk or "void" space contained within the lattice structure. This motivates an informal designation of $v(\mathbf{r})$ as a *void field*, a concept and its phenomenological consequences we develop in subsequent work.

3.2 The ellipticity and the shear modes

The traceless components of $\{\delta\theta_i\}$ are encoded in the pairwise differences. We define the ellipticity as the Euclidean norm on the three differences:

$$\mathcal{E}(\mathbf{r}) \;\equiv\; \sqrt{\sum_{i<j}\big(\delta\theta_i - \delta\theta_j\big)^2}. \tag{9}$$

The choice of squared differences here reflects that any linear-order anisotropy of $\{\delta\theta_i\}$ already contributes to v at first order, while genuinely *relative* departures from phase-locking appear at second order in the inner product computed below.

The algebraic identity

$$\mathcal{E}^2 \;=\; 3\sum_{i=1}^{3}(\delta\theta_i)^2 \;-\; \Big(\sum_{i=1}^{3}\delta\theta_i\Big)^2, \tag{10}$$

proved in Appendix B, expresses the ellipticity as the second moment of the $\{\delta\theta_i\}$ minus the squared trace.

3.3 Coherence function

Let the vacuum state of a single lattice cell be

$$|\psi_{\mathrm{vac}}\rangle \;=\; \frac{1}{\sqrt{3}}\sum_{i=1}^{3} e^{i\cdot 2\pi(i-1)/3}\,|i\rangle, \tag{11}$$

and let the perturbed state be the corresponding superposition with phases $\theta_i(\mathbf{r},t)$ from (1). The common time-dependent factor $e^{i\omega(\mathbf{r})t}$ cancels in the overlap, so we work with the static reduced states. The inner product is

$$\langle\psi_{\mathrm{vac}}|\psi(\mathbf{r})\rangle \;=\; \frac{1}{3}\sum_{i=1}^{3} e^{i\,\delta\theta_i(\mathbf{r})}, \tag{12}$$

the vacuum offsets $2\pi(i-1)/3$ cancelling exactly. Expanding to second order in $|\delta\theta_i| \ll 1$,

$$\mathrm{Re}\langle\psi_{\mathrm{vac}}|\psi\rangle \;\approx\; 1-\frac{1}{6}\sum_i(\delta\theta_i)^2, \tag{13}$$

$$\mathrm{Im}\langle\psi_{\mathrm{vac}}|\psi\rangle \;\approx\; \frac{1}{3}\sum_i \delta\theta_i \;=\; -v(\mathbf{r}). \tag{14}$$

The coherence function $\mathcal{C}(\mathbf{r}) = |\langle\psi_{\mathrm{vac}}|\psi\rangle|$ then takes the form

$$\mathcal{C}(\mathbf{r}) \;\approx\; 1 \;-\; \frac{\mathcal{E}^2(\mathbf{r})}{18}, \tag{15}$$

in which all v-dependence cancels exactly. The derivation is in Appendix A.

Equation (15) is the central structural result of this section. It states that the loss of overlap between the perturbed and vacuum states is controlled entirely by the shear modes. A uniform shift of all three phases changes the vacuum–perturbed inner product only at order v^2 in its imaginary part, but this contribution is removed by taking the modulus. The trace mode v measures the displacement of the mean phase relative to the vacuum, but it does not by itself reduce the relative coherence of the three orientations; only the spread $\mathcal{E}$ does.

3.4 Point-mass values and the model-independent ratio

For a point mass M at the origin, the integral (7) gives

$$v(r) \;=\; \frac{\alpha M}{r}. \tag{16}$$

For the minimal anisotropy configuration $\epsilon_1 = \epsilon$, $\epsilon_2 = -\epsilon$, $\epsilon_3 = 0$ (which satisfies the tracelessness constraint (4)), the calculation in Appendix C gives

$$\mathcal{E}(r) \;=\; \frac{\sqrt{6}\,\alpha\,\epsilon\,M}{r}. \tag{17}$$

Therefore

$$\boxed{\frac{v(\mathbf{r})}{\mathcal{E}(\mathbf{r})} \;=\; \frac{1}{\sqrt{6}\,\epsilon}} \tag{18}$$

for any point mass and at any distance, in the weak-response regime. This ratio is independent of M and r, and is therefore a structural prediction of the framework rather than a feature of any specific source configuration.

Remark 3 The structure of equations (7) and (9) makes manifest a clean separation of the static response into two independent sectors: an *isotropic* trace sector, characterized by v and obeying a Poisson equation, and a *traceless* shear sector, characterized by the ellipticity $\mathcal{E}$ and measuring relative coherence loss. The same source ρ_m populates both sectors. The next section shows that the gravitational interaction couples only to the trace sector; the shear sector therefore provides an independent channel that is, in principle, observable.

4 The Newtonian Limit

4.1 Matter–lattice energy coupling

To pass from the trace mode $v(\mathbf{r})$ to a mechanical force on a test mass, an additional input is required: a coupling between the test mass and the local state of the lattice.

Postulate 2 (Matter–lattice energy functional) A test mass m at position $\mathbf{r}$ has an energy $E_{\mathrm{dim}}(\mathbf{r})$ that depends on the local lattice state through a functional of the phases $\boldsymbol{\delta\theta}(\mathbf{r})$:

$$E_{\mathrm{dim}}(\mathbf{r}) \;=\; E_{\mathrm{dim}}\big[\delta\theta_1(\mathbf{r}), \delta\theta_2(\mathbf{r}), \delta\theta_3(\mathbf{r})\big]. \tag{19}$$

We further assume that E_{dim} is analytic at the vacuum configuration $\boldsymbol{\delta\theta} = \mathbf{0}$, where it takes the value $\mu\mathcal{D}_0\, m$ with $\mathcal{D}_0 = 3$ and $\mu > 0$ a coupling with dimensions L^2T^{-2}.

The parameters μ and $\mathcal{D}_0$ are kinematic: μ sets an energy scale and $\mathcal{D}_0$ counts the three orientations as the effective dimensional support of the vacuum. If $\mu\mathcal{D}_0$ is identified with the rest-energy scale c^2, then $\mu \simeq c^2/3$; the Newtonian limit fixes only the product $\mu\lambda\alpha$ that appears in Newton's constant.

4.2 Symmetry-constrained leading correction

Proposition 2 (Linearity in the trace mode $v(\mathbf{r})$) *Under Postulate 2, the first-order Taylor expansion of E_{dim} about the vacuum takes the form*

$$E_{\mathrm{dim}}(\mathbf{r}) \;=\; \mu m\,\mathcal{D}(\mathbf{r}) \;+\; \mathcal{O}\big((\delta\theta_i)^2\big), \qquad \mathcal{D}(\mathbf{r}) \;\equiv\; \mathcal{D}_0 - \lambda\, v(\mathbf{r}), \tag{20}$$

for some real $\lambda = 3\beta_0/\mu$, where β_0 is the common value of the first partial derivatives of E_{dim} at the vacuum.

Proof Taylor-expand E_{dim} about $\boldsymbol{\delta\theta} = \mathbf{0}$:

$$\begin{aligned} E_{\mathrm{dim}}(\mathbf{r}) \;&=\; \mu\mathcal{D}_0 m \;+\; \sum_{i=1}^{3} \beta_i\, \delta\theta_i(\mathbf{r}) \;+\; \mathcal{O}\big((\delta\theta_i)^2\big), \\ \beta_i \;&\equiv\; \left.\frac{\partial E_{\mathrm{dim}}}{\partial(\delta\theta_i)}\right|_{\mathbf{0}}. \end{aligned} \tag{21}$$

The vacuum state (11) is invariant under the cyclic permutation $\sigma\colon 1 \mapsto 2 \mapsto 3 \mapsto 1$ up to an overall phase that does not affect E_{dim}. The functional E_{dim} is therefore invariant under σ at the vacuum point, which forces

$$\beta_1 \;=\; \beta_2 \;=\; \beta_3 \;\equiv\; \beta_0\, m, \tag{22}$$

where the factor of m is required by the weak equivalence principle in the test-mass limit [18]. Therefore

$$\sum_{i=1}^{3} \beta_i\, \delta\theta_i \;=\; \beta_0\, m \sum_{i=1}^{3} \delta\theta_i \;=\; -3\beta_0\, m\, v(\mathbf{r}), \tag{23}$$

the last equality using definition (6). Setting $\lambda = 3\beta_0/\mu$ gives (20). □

Remark 4 Proposition 2 is a representation-theoretic fact: under the cyclic group $\mathbb{Z}_3$, the three-dimensional space of phase perturbations decomposes into a one-dimensional trivial representation (the trace mode v) and a two-dimensional non-trivial representation (the shear modes). A scalar functional invariant under $\mathbb{Z}_3$ at the vacuum has a leading-order coupling only to the trivial representation. The linearity of $\mathcal{D}$ in v is therefore not an ansatz; it is the leading-order content of $\mathbb{Z}_3$-equivariance.

The second-order corrections to (20) are suppressed by additional powers of $|\delta\theta_i| \ll 1$, the same small parameter that controls the validity of the phase-response ansatz (2).

4.3 Force law and Newton's constant

The force on the test mass follows by standard mechanics from the energy functional (20):

$$\mathbf{F}(\mathbf{r}) \;=\; -\nabla E_{\mathrm{dim}}(\mathbf{r}) \;=\; \mu\lambda m\, \nabla v(\mathbf{r}). \tag{24}$$

Defining the gravitational potential

$$\Phi_{\mathrm{grav}}(\mathbf{r}) \;\equiv\; -\kappa_g\, v(\mathbf{r}), \qquad \kappa_g \;\equiv\; \mu\lambda, \tag{25}$$

the force takes the standard form $\mathbf{F} = -m\nabla\Phi_{\mathrm{grav}}$. For a point mass M at the origin, $v(r) = \alpha M/r$ from (16), and

$$\Phi_{\mathrm{grav}}(r) \;=\; -\frac{\mu\lambda\alpha M}{r}. \tag{26}$$

Comparison with the Newtonian potential $\Phi_N = -GM/r$ gives

$$\boxed{\,G \;=\; \mu\lambda\alpha\,} \tag{27}$$

and the inverse-square law

$$\mathbf{F} \;=\; -\frac{GmM}{r^2}\,\hat{\mathbf{r}} \tag{28}$$

in the static, weak-response regime.

4.4 Consistency of the frequency shift

The common-mode frequency shift (3) does not enter the force-law derivation; the gradient of E_{dim} involves only the static phase perturbations $\delta\theta_i$. The parameter ζ_P is therefore a free coupling at the level of the force law.

A consistency condition fixes ζ_P uniquely. The phase-response ansatz describes a single physical situation in which the same mass distribution ρ_m sources both the gravitational potential (25) and the frequency shift (3). Requiring the latter to reproduce the standard weak-field clock-rate shift,

$$\frac{\omega(\mathbf{r}) - \omega_0}{\omega_0} \;=\; \frac{\Phi_N(\mathbf{r})}{c^2} \tag{29}$$

[17, 19, 20], gives

$$\boxed{\zeta_P \;=\; \frac{G}{c^2} \;=\; \frac{\mu\lambda\alpha}{c^2}.} \tag{30}$$

This is a derived consistency relation among the framework parameters; the force law and the clock-rate shift are governed by the same Newton constant.

4.5 Logical structure

Three inputs determine the Newtonian limit. The phase-response ansatz (Postulate 1) fixes the spatial response of the lattice phases to mass. The matter–lattice coupling (Postulate 2) gives E_{dim} as a functional of the lattice state. The three-fold vacuum symmetry constrains the leading-order form of E_{dim}. The output is a Poisson equation for v, a force law linear in ∇v, the identification $G = \mu\lambda\alpha$, and the consistency relation $\zeta_P = G/c^2$.

The framework does not derive the response kernel $1/|\mathbf{r} - \mathbf{r}'|$ from microscopic lattice dynamics. Doing so requires a specific Hamiltonian whose continuum limit reproduces the assumed kernel. This is the content of the microscopic completion program discussed in Section 6.

5 Predictions: Correlated Anisotropic Phase Response

The framework constructed in Sections 2–4 reproduces the Newtonian limit by construction and is, to that extent, indistinguishable from Newtonian gravity. Its physical content lies in the shear sector, which is populated by the same source but does not contribute to the gravitational force. The shear sector predicts a specific angular signature with magnitude tied to the local gravitational potential, mapping onto the same class of observables searched for in precision tests of Lorentz invariance and the

equivalence principle. This section makes that mapping explicit, derives quantitative bounds on the anisotropy parameter from existing experimental data, and proposes a direct test of the framework's defining prediction: that the anisotropic signal scales linearly with the local gravitational potential and vanishes in the absence of nearby gravitating sources.

5.1 The fixed-ratio prediction

For the minimal anisotropy configuration of Section 3, equation (18) states

$$\frac{v(\mathbf{r})}{\mathcal{E}(\mathbf{r})} = \frac{1}{\sqrt{6}\,\epsilon}, \tag{31}$$

independent of source mass and distance. Combining with (25) and (27), and using the identification $\mu\mathcal{D}_0 \simeq c^2$ from Section 4,

$$\mathcal{E}(\mathbf{r}) = \frac{3\sqrt{6}\,\epsilon}{\lambda} \cdot \frac{|\Phi_{\text{grav}}(\mathbf{r})|}{c^2}. \tag{32}$$

The ellipticity is a dimensionless measure of anisotropic phase coherence loss whose magnitude is fixed by the product of the anisotropy parameter, the inverse trace–dimensionality coupling, and the local gravitational potential in units of c^2. The dependence on λ is retained explicitly: the shear-sector predictions constrain the combination ϵ/λ rather than ϵ alone, with $\lambda = \mathcal{O}(1)$ the economical working assumption.

5.2 Angular structure: the quadrupole tensor

The ellipticity $\mathcal{E}$ is a scalar magnitude. The underlying anisotropic structure is encoded in the orientation-resolved phase shifts $\delta\theta_i(\mathbf{r})$, which under the tracelessness condition (4) form a traceless three-component object indexed by the internal lattice orientations $i \in \{1, 2, 3\}$. Two additional assumptions are required to extract observable angular structure from this internal object. We state them explicitly and discuss their status.

Tetrad assumption.

We assume that the three lattice orientations are projected onto three orthogonal directions of a microscopic spatial frame $\{\hat{n}_a\}_{a=1,2,3}$ through a tetrad or vielbein structure. This is an assumption beyond Postulates 1 and 2; the matter–lattice coupling specified in Postulate 2 acts on internal phases and does not by itself fix how those phases are projected onto spatial directions. A microscopic completion of the framework must supply this projection. In its absence we take the simplest case: an orthogonal projection of the three internal orientations onto three spatial axes, with no mixing between internal and spatial indices.

Eigenframe orientation.

The tetrad assumption fixes the eigenvalue structure of the spatial shear tensor but not the orientation of the eigenframe $\{\hat{n}_a\}$ in physical space. Three logically distinct possibilities exist:

(i) Fixed cosmic frame. The eigenframe is determined by a preferred direction of spacetime (e.g., the cosmic microwave background rest frame). This reintroduces a global preferred direction at the macroscopic level, in contradiction with the rotational isotropy condition that motivated the tracelessness constraint (4). This possibility is inconsistent with the framework's structural assumptions.

(ii) Matter-locked frame. The eigenframe is co-moving with each test mass. Then the anisotropy of the lattice and the orientation of the measurement system rotate together, and no observable signal arises. This possibility renders the shear sector empirically empty.

(iii) Source-locked frame. The eigenframe at position $\mathbf{r}$ is determined by the local gradient structure of the gravitating sources, with the principal eigendirection aligned with $\nabla\Phi_{\rm grav}(\mathbf{r})$ of the dominant source. The eigenframe is therefore a smooth function of position, set by the source configuration rather than by a fixed cosmic background or by the test system.

Of the three, only possibility (iii) is structurally consistent and observationally non-trivial. We adopt it. For a laboratory on Earth, the dominant gradient is Earth-radial, and the principal eigendirection points vertically. The contribution from the Sun's potential adds a smaller component aligned with the Sun direction; analogous contributions from the Galactic centre and other distant sources are negligible in magnitude over the present discussion. The total shear tensor at the lab is the sum of contributions from each source, each with its own eigenframe and magnitude.

The spatial shear tensor.

Under these assumptions, the asymmetry parameters $(\epsilon_1, \epsilon_2, \epsilon_3)$ from a single dominant source produce the symmetric traceless rank-2 tensor

$$S_{ij}(\mathbf{r}) \;=\; \frac{|\Phi_{\rm grav}(\mathbf{r})|}{c^2} \sum_{a=1}^{3} \epsilon_a \, \hat{n}_a^i \, \hat{n}_a^j, \qquad \mathrm{Tr}\, S = 0, \tag{33}$$

with the eigenframe $\{\hat{n}_a\}$ set by the source geometry as discussed above. The tracelessness condition (4) reduces the eigenvalue content from three to two independent parameters. Combined with the eigenframe orientation being fixed by physics rather than free, the framework predicts a quadrupole with two free parameters, in contrast to the five independent components of a generic Lorentz-violating traceless symmetric tensor. This restriction is a structural feature of the framework and appears below as the cleanest distinguishing prediction against generic Lorentz-violation parametrizations.

Angular signature.

Acting on a test system with internal frequency ω_0 and orientation $\hat{m}$ relative to the source-locked eigenframe, the shear sector produces a direction-dependent fractional

frequency shift

$$\frac{\delta\omega(\hat{m})}{\omega_0} \sim S_{ij}\, m^i m^j, \tag{34}$$

which has the spherical-harmonic content of a generic $Y_{2,m}$ quadrupole. For the minimal eigenvalue configuration $\epsilon_1 = \epsilon$, $\epsilon_2 = -\epsilon$, $\epsilon_3 = 0$, the angular pattern reduces to the $\mathrm{Re}\, Y_{2,2}$ form with a fixed relation between $Y_{2,0}$ and $Y_{2,\pm 2}$ amplitudes; more general traceless eigenvalue configurations give the same $Y_{2,m}$ content with shifted relative amplitudes. In every case the angular pattern is parity-even, rotates at twice the lab–eigenframe relative angular frequency, and is restricted to the two-parameter family identified above.

5.3 Bounds from existing precision tests

The Standard-Model Extension (SME) [25, 26] provides a framework-independent parametrization of Lorentz-violating couplings, including traceless symmetric rank-2 tensors in the photon, matter, and gravity sectors. The shear-sector prediction (33) maps onto specific SME coefficient combinations once the microscopic matter–lattice projection is specified; while that specification is beyond the scope of the present effective treatment, existing experimental bounds on the relevant coefficient classes constrain ϵ/λ at the orders of magnitude estimated below.

We quote bounds on the combination ϵ/λ throughout this subsection; the corresponding bound on ϵ alone assumes the working hypothesis $\lambda = \mathcal{O}(1)$ identified in Section 4. The framework parameters ϵ and λ are in principle independently measurable: λ enters the Newton constant $G = \mu\lambda\alpha$ and is therefore probed by tests of the gravitational sector, while ϵ controls only the anisotropic shear channel. A combined analysis of gravitational and shear-sector measurements would separate the two parameters.

Optical cavity tests.

Modern Michelson–Morley-type experiments using rotating cryogenic sapphire cavities and Fabry–Perot resonators bound the symmetric traceless coefficients $(\tilde{\kappa}_{e-})_{jk}$ of the photon sector at the 10^{-18} level [27]. If the matter–lattice coupling projects through electromagnetic transitions, the shear sector contributes to these coefficients at the level $(\epsilon/\lambda)\cdot|\Phi_\odot|/c^2$. Taking the Sun's potential at Earth, $|\Phi_\odot|/c^2 \simeq 10^{-8}$, this gives

$$\left|\frac{\epsilon}{\lambda}\right| \lesssim 10^{-10} \qquad \text{(cavity bound)}. \tag{35}$$

Trapped-ion and atomic-spectroscopy precision tests.

Searches for Lorentz violation in the electronic structure of $^{40}\mathrm{Ca}^+$, $^{171}\mathrm{Yb}^+$, and $^{162}\mathrm{Dy}$ bound traceless matter-sector coefficients at the 10^{-19} to 10^{-21} level [21, 28–30]. With an analogous projection through the electronic matter sector,

$$\left|\frac{\epsilon}{\lambda}\right| \lesssim 10^{-11} \text{ to } 10^{-13} \qquad \text{(clock and spectroscopy bound)}, \tag{36}$$

the range reflecting both the experimental spread and the unspecified projection coefficients.

Lunar laser ranging and atom interferometry.

The SME gravity-sector coefficients $\bar{s}_{JK}$, which directly parametrize potential-dependent anisotropy of the type predicted by (33), are bounded at the 10^{-9} to 10^{-11} level by lunar laser ranging [31] and atom-interferometric tests of the equivalence principle [32]. These give

$$\left|\frac{\epsilon}{\lambda}\right| \lesssim 10^{-9} \text{ to } 10^{-11} \qquad \text{(gravity-sector bound)}. \tag{37}$$

Combined assessment.

The bounds (35)–(37) constrain ϵ/λ at the level 10^{-9} to 10^{-13} across all currently relevant experimental channels. The conservative bound $|\epsilon/\lambda| \lesssim 10^{-10}$ holds for any projection consistent with existing optical cavity results and the Sun's gravitational potential at Earth. Under $\lambda = \mathcal{O}(1)$ this is also the bound on ϵ alone.

5.4 A direct test: gravitational potential scaling

Equation (32) makes a distinctive prediction that separates the framework from generic Lorentz violation: the magnitude of the anisotropic signal is proportional to the local gravitational potential $|\Phi_{\text{grav}}|/c^2$, not a fixed coefficient. Standard SME treatments take the Lorentz-violating coefficients as constants of nature; the shear sector predicts they should vary measurably with position in any gravitational potential gradient.

This is the framework's most direct testable handle. Three feasible comparisons probe the predicted scaling using either existing data or near-future precision measurements.

Annual modulation of cavity and clock anisotropy.

Earth's orbital eccentricity $e \simeq 0.0167$ modulates the Sun's potential at Earth by

$$\frac{\Delta|\Phi_\odot|}{c^2} \simeq 2e \cdot \frac{|\Phi_\odot|}{c^2} \simeq 3 \times 10^{-10} \tag{38}$$

over the year. The framework predicts a corresponding annual modulation of the anisotropic signal at the level $3 \times 10^{-10} \cdot (\epsilon/\lambda)$. For $|\epsilon/\lambda|$ at the current cavity bound 10^{-10}, the predicted modulation is $\sim 3 \times 10^{-20}$, below current single-experiment sensitivity but accessible to next-generation optical clock networks [20, 29]. A reanalysis of existing multi-year cavity-anisotropy data for annual modulation of the $\bar{s}_{JK}$-type coefficient pattern is already feasible.

Altitude variation.

Vertical baselines between ground-based and mountain-altitude clock comparisons access fractional potential differences $\Delta|\Phi_\oplus|/c^2 \sim 10^{-12}$ to 10^{-13}. The millimetre-scale gravitational redshift demonstrated in [20] establishes the relevant sensitivity

floor. A controlled altitude-variation measurement of anisotropic signal magnitude, comparing the predicted $\Delta\mathcal{E} \propto \Delta|\Phi_\oplus|$ against the observed change, would test the scaling directly.

Pulsar timing in eccentric binaries.

Companions of millisecond pulsars in eccentric binary orbits sit in gravitational potentials $|\Phi|/c^2 \sim 10^{-6}$, two orders of magnitude above the Sun's potential at Earth and four orders of magnitude above Earth's surface potential. While the relevant atomic and nuclear systems differ from laboratory clocks, the framework predicts orientation-dependent timing residuals enhanced by a factor of $\sim 10^2$ over solar-system analogues. The high-precision timing programs at FAST, MeerKAT, and the SKA pulsar timing array [33] are sensitive to such residuals at fractional levels approaching those required to test (32) at the current bound on ϵ/λ.

What a positive or null result would mean.

A detection of anisotropic frequency shifts whose magnitude fails to scale linearly with $|\Phi|/c^2$ across two or more independent comparisons would rule out the framework, even if generic LIV-type signals are detected. A detection of the predicted scaling, with the source-locked eigenframe orientation and the restricted two-parameter eigenvalue content identified in Section 5.2, would constrain the microscopic origin of the asymmetry parameters in ways that no fixed-coefficient parametrization can match. The framework therefore makes a specifically falsifiable prediction beyond the generic LIV landscape.

5.5 Distinguishing features

The shear-sector prediction has three structural features that separate it from generic Lorentz or rotational symmetry violations. We list them in order of their strength as discriminators.

(1) Strict correlation with gravitating sources.

Generic SME parametrizations treat Lorentz-violating coefficients as constants of nature, present whether or not gravitating mass is nearby. The shear sector predicts that the anisotropic signal vanishes in the absence of any local gravitating source producing a non-uniform potential: no mass gradient, no anisotropy. A detection of an anisotropic phase response in a configuration with no nearby local mass distribution would rule out the framework. Conversely, a fixed-coefficient LIV detection that does not vanish as sources are removed cannot be the framework's signal. This is the structural prediction that no constant-coefficient parametrization can reproduce.

(2) Linear scaling with the gravitational potential.

Among predictions tied to gravitating sources, the framework makes a specific quantitative claim: the magnitude of the anisotropic signal is proportional to the local $|\Phi_{\rm grav}|/c^2$. This is the prediction tested by the scaling comparisons of Section 5.4 and

the most direct experimental contact between the framework and existing precision-test programs. A measurement consistent with the framework's strict correlation but failing the linear scaling would also rule it out.

(3) Restricted parameter content of the quadrupole.

A generic traceless symmetric rank-2 tensor in three dimensions has five independent components: two eigenvalue parameters and three Euler angles for the eigenframe orientation. The framework predicts a quadrupole with two eigenvalue parameters and an eigenframe orientation fixed by the local source geometry rather than independent. Both the eigenframe orientation and the eigenvalue pattern are therefore testable: a measurement of the relative amplitudes of the $Y_{2,0}$ and $Y_{2,\pm 2}$ spherical-harmonic components, together with the eigenframe orientation in space, would discriminate the framework from broader SME parametrizations that leave all five components free.

6 Scope and Open Problems

The framework presented above is a kinematic effective description. We close by stating what it does not address and identifying the principal open problems for a deeper completion.

6.1 Domain of validity

The construction is valid only in the static, weak-response regime (5). The linear phase ansatz and the first-order Taylor expansion of $E_{\rm dim}$ break down at the same scale $\alpha M/r \sim 1$, so the framework is mutually consistent within its domain but provides no access to strong-field gravity, dynamics, or Lorentz-covariant extensions. A relativistic generalization would require replacing the static Green-function ansatz with a covariant propagator and lifting the matter–lattice functional to a Lorentz scalar; we do not undertake this here.

6.2 The microscopic Hamiltonian

Postulate 2 specifies an effective functional form for $E_{\rm dim}$ without committing to a microscopic Hamiltonian. A natural candidate is the cosine operator

$$\hat{V} \;=\; \frac{1}{3}\sum_{i=1}^{3}\cos(\theta_i(\mathbf{r}))\,\hat{I}, \qquad H_{\rm int} \;=\; E_0\,\hat{V}. \tag{39}$$

Appendix D establishes that this candidate fails: its first-order expectation value couples to the shear mode $(\delta\theta_3 - \delta\theta_2)$ rather than to the trace mode, and the trace-mode coupling required by Proposition 2 is identically zero. The same conclusion holds for any analytic function of $\sum_i \cos\theta_i$. A valid microscopic completion of Postulate 2 must therefore be non-linear in the phase variables in a way that breaks the $\sin\theta_{0i}$ selection rule, or must couple to the trace mode through a tetrad or vielbein structure that projects the three orientations onto the spatial frame.

6.3 Planck-scale origin of the asymmetry parameters

If the ϵ_i are identified with the eigenvalues of a macroscopic deviatoric stress tensor, they vanish for any spherically symmetric mass distribution. Under this identification the shear sector predicts no signal from isotropic sources, and the structural prediction of Section 5 is empty. The framework therefore requires that the ϵ_i arise from a microscopic mechanism that distinguishes the three orientations independently of the macroscopic stress distribution, yet dynamically aligns the resulting spatial eigenframe with the local gravitational gradient (as adopted phenomenologically in Section 5.2). A spinor or vielbein projection of the energy-momentum tensor onto the lattice orientations is a plausible mechanism, but its construction lies outside the present effective treatment.

6.4 Dynamical realization of coherence loss

The trace mode and ellipticity are static quantities. Under Postulate 1 as stated, the perturbed state is an exact eigenstate of the local Hamiltonian, the cyclic work over one oscillation period vanishes identically, and there is no genuine dissipative loss of coherence [22]. A dynamical interpretation of the ellipticity as a measure of irreversible phase-space area requires the three orientations to oscillate at distinct frequencies. The minimal modification is to replace (3) by

$$\omega_i(\mathbf{r}) \;=\; \omega_0\left(1+\frac{\Phi_{\mathrm{grav}}(\mathbf{r})}{c^2}+\gamma\,\epsilon_i\right), \tag{40}$$

with γ a dimensionless coupling. The trace mode $v(\mathbf{r})$ and the Newtonian force law are unaffected by this modification, but the cyclic work acquires a contribution $W_{\mathrm{cycle}} \propto \mathcal{E}^2$. Deriving γ from a microscopic theory and verifying the energy budget is left to future work.

6.5 Relation to existing programs

The framework is closest in spirit to fundamental-decoherence proposals [14–16], in which gravitational sources modify the dynamics of quantum systems. It differs in predicting a structural correlation between the gravitational potential and an anisotropic, rather than universal, coherence loss. The framework is orthogonal to entropic and thermodynamic emergent-gravity proposals [2, 3, 5], which derive Einstein-type equations from horizon thermodynamics rather than from an explicit microscopic substrate. The discrete-geometry programs [7–11] provide candidate microscopic substrates whose continuum limits could in principle realize the phase-response ansatz of Postulate 1; identifying a specific discrete model whose long-wavelength limit reproduces the trace-mode Poisson equation (8) is an open problem.

We have presented an effective scalar order-parameter framework in which mass induces static phase shifts in three internal orientations of a hypothetical microscopic lattice. The static response decomposes by linear algebra into an isotropic trace mode v and traceless shear modes encoded in an ellipticity $\mathcal{E}$. The trace mode obeys a Poisson equation by construction; a matter–lattice energy functional, constrained at leading

order by the three-fold vacuum symmetry, gives a Newtonian force law with $G = \mu\lambda\alpha$ in terms of three effective Planck-scale parameters. A consistency condition with the standard weak-field clock-rate shift fixes the frequency–mass coupling as $\zeta_P = G/c^2$.

The framework is offered as a kinematic building block. The Newtonian limit is recovered, not derived from microscopic dynamics, and the recovery serves only as a consistency check. The genuine content lies in the shear sector, which predicts an anisotropic phase response strictly correlated with the local gravitational potential. Under the phenomenological assumption of a source-locked spatial projection, the framework yields quantitative bounds on the parameter ratio $|\epsilon/\lambda| \lesssim 10^{-10}$ using existing precision frequency-isotropy data. Most importantly, it makes a uniquely falsifiable prediction separating it from generic Lorentz violation: the magnitude of the anisotropic signal must scale linearly with $|\Phi_{\rm grav}|/c^2$ and is sourced by local gravitating mass distributions.

Three open problems—a microscopic Hamiltonian realizing the required trace-mode coupling, the Planck-scale origin of the anisotropy parameters, and a dynamical mechanism for coherence loss—are identified as targets for a deeper completion of the framework. We do not view the present construction as a candidate theory of gravity; we view it as a kinematic representation of one possible structural relation between gravitational attraction and anisotropic coherence loss, to be either supported or excluded by future experimental work and by the success or failure of a microscopic completion.

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Declarations

Conflict of interest/Competing interests: The authors declare no competing interests.

Appendix A Inner Product and Coherence Function

With $\theta_i = \omega(\mathbf{r})t + 2\pi(i-1)/3 + \delta\theta_i(\mathbf{r})$, the perturbed cell state is

$$|\psi(\mathbf{r},t)\rangle \;=\; \frac{e^{i\omega(\mathbf{r})t}}{\sqrt{3}} \sum_{i=1}^{3} e^{i(2\pi(i-1)/3+\delta\theta_i)}\,|i\rangle. \tag{A1}$$

Using the orthonormality of $\{|i\rangle\}$, the inner product with the vacuum state (11) is

$$\begin{aligned}\langle\psi_{\mathrm{vac}}|\psi(\mathbf{r},t)\rangle \;&=\; \frac{e^{i\omega(\mathbf{r})t}}{3} \sum_{i=1}^{3} e^{-i\,2\pi(i-1)/3}\, e^{i(2\pi(i-1)/3+\delta\theta_i)} \\ &=\; \frac{e^{i\omega(\mathbf{r})t}}{3} \sum_{i=1}^{3} e^{i\,\delta\theta_i}. \end{aligned}\tag{A2}$$

The vacuum offsets cancel exactly. The common time factor contributes a unit-modulus prefactor that cancels in $\mathcal{C} = |\langle\psi_{\mathrm{vac}}|\psi\rangle|$; equation (12) follows.

Expanding $e^{i\delta\theta_i} \approx 1 + i\delta\theta_i - \frac{1}{2}(\delta\theta_i)^2$ and averaging:

$$\mathrm{Re}\langle\psi_{\mathrm{vac}}|\psi\rangle \;\approx\; 1 - \frac{1}{6}\sum_i (\delta\theta_i)^2, \tag{A3}$$

$$\mathrm{Im}\langle\psi_{\mathrm{vac}}|\psi\rangle \;\approx\; \frac{1}{3}\sum_i \delta\theta_i \;=\; -v. \tag{A4}$$

The squared modulus is

$$\mathcal{C}^2 \;\approx\; 1 - \frac{1}{3}\sum_i (\delta\theta_i)^2 + \frac{1}{9}\Big(\sum_i \delta\theta_i\Big)^2. \tag{A5}$$

Using $\sum_i \delta\theta_i = -3v$ and the ellipticity identity (10) to write $\sum_i (\delta\theta_i)^2 = (\mathcal{E}^2 + 9v^2)/3$,

$$\mathcal{C}^2 \;\approx\; 1 - \frac{\mathcal{E}^2 + 9v^2}{9} + \frac{9v^2}{9} \;=\; 1 - \frac{\mathcal{E}^2}{9}. \tag{A6}$$

The v^2-terms cancel exactly. Taking the square root via $(1-x)^{1/2} \approx 1 - x/2$ for small x gives $\mathcal{C} \approx 1 - \mathcal{E}^2/18$, equation (15).

Appendix B Ellipticity Identity

We prove (10). Expanding the three squared pairwise differences,

$$\begin{aligned}\mathcal{E}^2 \;&=\; (\delta\theta_1 - \delta\theta_2)^2 + (\delta\theta_2 - \delta\theta_3)^2 + (\delta\theta_3 - \delta\theta_1)^2 \\ &=\; 2\sum_i (\delta\theta_i)^2 - 2(\delta\theta_1\delta\theta_2 + \delta\theta_2\delta\theta_3 + \delta\theta_3\delta\theta_1). \end{aligned}\tag{B7}$$

Eliminating the cross-terms via

$$\Big(\sum_i \delta\theta_i\Big)^2 \;=\; \sum_i (\delta\theta_i)^2 + 2(\delta\theta_1\delta\theta_2 + \delta\theta_2\delta\theta_3 + \delta\theta_3\delta\theta_1) \tag{B8}$$

yields $\mathcal{E}^2 = 3\sum_i(\delta\theta_i)^2 - (\sum_i \delta\theta_i)^2$. □

Appendix C Point-Mass Ellipticity

For a point source M at the origin with the minimal anisotropy configuration $\epsilon_1 = \epsilon$, $\epsilon_2 = -\epsilon$, $\epsilon_3 = 0$, the phase-shifts (2) become

$$\delta\theta_1 \;=\; -\frac{\alpha M(1+\epsilon)}{r}, \quad \delta\theta_2 \;=\; -\frac{\alpha M(1-\epsilon)}{r}, \quad \delta\theta_3 \;=\; -\frac{\alpha M}{r}. \tag{C9}$$

The three pairwise differences are

$$\begin{aligned} \delta\theta_1 - \delta\theta_2 \;&=\; -\frac{2\epsilon\alpha M}{r}, \\ \delta\theta_2 - \delta\theta_3 \;&=\; \frac{\epsilon\alpha M}{r}, \\ \delta\theta_3 - \delta\theta_1 \;&=\; \frac{\epsilon\alpha M}{r}. \end{aligned} \tag{C10}$$

Substituting into (9),

$$\mathcal{E}^2 \;=\; \left(\frac{\alpha M}{r}\right)^2 (4\epsilon^2 + \epsilon^2 + \epsilon^2) \;=\; \frac{6\epsilon^2\alpha^2 M^2}{r^2}, \tag{C11}$$

giving $\mathcal{E}(r) = \sqrt{6}\,\alpha\,\epsilon\,M/r$, equation (17). Combined with $v(r) = \alpha M/r$ from (16), $v/\mathcal{E} = 1/(\sqrt{6}\,\epsilon)$ independent of M and r.

Appendix D Constraint on the Microscopic Hamiltonian

We establish that the natural cosine Hamiltonian (39) does not realize Postulate 2. The result identifies the structural constraint on a valid microscopic completion.

Proposition 3 *Let the lattice phases be $\theta_i(\mathbf{r}) = 2\pi(i-1)/3 + \delta\theta_i(\mathbf{r})$ with $|\delta\theta_i| \ll 1$, and let $v(\mathbf{r})$ be defined by (6). The first-order Taylor expansion of $E_{\mathrm{dim}} = \langle\psi_{\mathrm{vac}}|H_{\mathrm{int}}|\psi_{\mathrm{vac}}\rangle$ in $\{\delta\theta_i\}$, with $H_{\mathrm{int}} = E_0\hat{V}$ and $\hat{V} = \frac{1}{3}\sum_i \cos(\theta_i(\mathbf{r}))\hat{I}$, is*

$$\Delta E^{(1)} \;=\; \frac{\sqrt{3}}{2}\,\mu m\big(\delta\theta_3(\mathbf{r}) - \delta\theta_2(\mathbf{r})\big). \tag{D12}$$

The coefficient of any function of v alone vanishes identically.

Proof The vacuum phases are $\theta_{0,1} = 0$, $\theta_{0,2} = 2\pi/3$, $\theta_{0,3} = 4\pi/3$, giving $\sin\theta_{0,1} = 0$, $\sin\theta_{0,2} = \sqrt{3}/2$, $\sin\theta_{0,3} = -\sqrt{3}/2$, and $\sum_i \cos\theta_{0,i} = 0$. Expanding $\cos(\theta_{0,i} + \delta\theta_i) \approx \cos\theta_{0,i} - \sin\theta_{0,i}\,\delta\theta_i$ to first order, the zeroth-order contribution vanishes by the three-fold symmetry. Setting the macroscopic energy scale $E_0 = 3\mu m$ to match Postulate 2, the first-order correction is

$$\begin{aligned} \Delta E^{(1)} &= -\mu m \sum_{i=1}^{3} \sin\theta_{0,i}\,\delta\theta_i \\ &= -\mu m\Big(0\cdot\delta\theta_1 + \tfrac{\sqrt{3}}{2}\delta\theta_2 - \tfrac{\sqrt{3}}{2}\delta\theta_3\Big) \\ &= \tfrac{\sqrt{3}}{2}\mu m(\delta\theta_3 - \delta\theta_2). \end{aligned} \tag{D13}$$

This is (D12). The combination $\delta\theta_3 - \delta\theta_2$ is the difference of a pair of phases and is therefore traceless: it is orthogonal in $\mathbb{R}^3$ to the trace mode $\sum_i \delta\theta_i$, and no function of $v = -(1/3)\sum_i \delta\theta_i$ alone can reproduce it. □

Remark 5 Proposition 3 extends to any analytic function $f\big(\sum_i \cos\theta_i\big)$: the zeroth-order cancellation of $\sum_i \cos\theta_{0,i}$ is a consequence of three-fold symmetry, and the first-order correction always selects the shear modes through the antisymmetry of $\sin\theta_{0,i}$ under reflection $i \leftrightarrow -i$. A microscopic Hamiltonian realizing Postulate 2 must therefore (i) be non-linear in the phase variables in a way that breaks this selection, or (ii) couple to the trace mode through an explicit tetrad or vielbein projection of the three orientations onto the spatial frame. Constructing such a Hamiltonian is left to future work.

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